

2022

HIGHER
SCHOOL
CERTIFICATE
TRIAL EXAMINATION

Mathematics Advanced

General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- In Questions 11–32 show relevant mathematical reasoning and/or calculations
- Write your Student ID below and on page 7

Total marks:
100

Section I – 10 marks (pages 1–6)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 7–27)

- Attempt Questions 11–32
- Allow about 2 hours and 45 minutes for this section

STUDENT ID _____



THIS PAPER CANNOT BE RELEASED IN PUBLIC UNTIL AFTER 26th AUGUST 2022

This paper is used with the understanding that it has a Security Period. ©Total Education Centre

Section I**10 marks****Attempt questions 1-10****Allow about 15 minutes for this section**

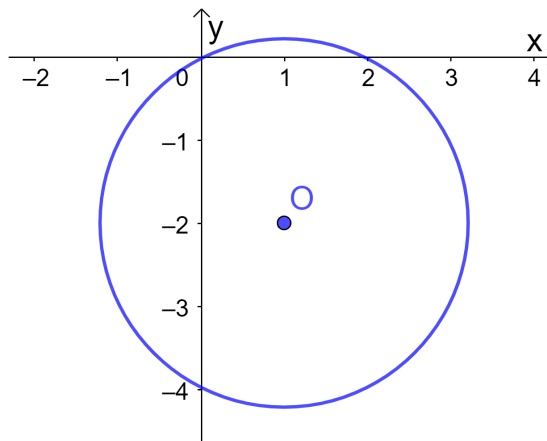
Use the multiple-choice answer sheet for Questions 1-10.

- 1 The first term of an arithmetic sequence is 12. The fifteenth term of the same sequence is 67.

What is the common difference of the sequence?

- A. $\frac{11}{3}$
B. $\frac{55}{14}$
C. 3
D. $\frac{45}{14}$

- 2 In the diagram, the graph of the circle with centre $O(1, -2)$ is shown.



Which of the following gives the equation of the circle?

- A. $(x+1)^2 + (y-2)^2 = 25$
B. $(x-1)^2 + (y+2)^2 = 25$
C. $(x-1)^2 + (y+2)^2 = 5$
D. $(x+1)^2 + (y-2)^2 = 5$

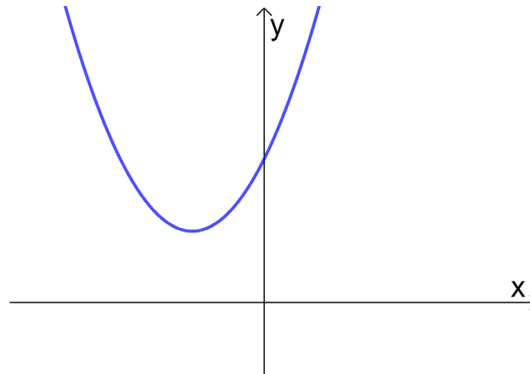
3 How many stationary points does the function $f(x) = 4x^3 - 6x^2 + 3x$ have?

- A. 0
- B. 1
- C. 2
- D. 3

4 Which of the following is equivalent to $\sec^2 x - \sin^2 x - \cos^2 x$?

- A. $\tan^2 x + 1$
- B. $1 - \tan^2 x$
- C. $\tan^2 x - 1$
- D. $\tan^2 x$

5 The diagram shows the graph of $y = x^2 + bx + 2$.



Which of the following could be the value of b ?

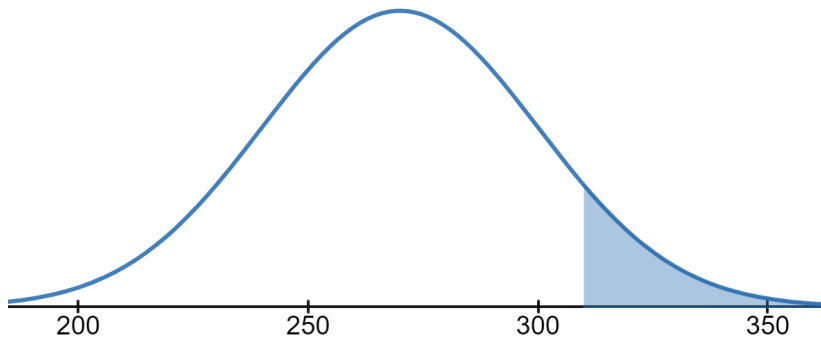
- A. 2
- B. -2
- C. 3
- D. -3

- 6 Given the function $y = \frac{2}{1-e^x}$, which expression is equal to $\frac{dy}{dx}$?
- A. $\frac{2e^x}{(1-e^x)^2}$
- B. $\frac{-2e^x}{1-e^x}$
- C. $\frac{-2}{(1-e^x)^2}$
- D. $\frac{e^x}{(1-e^x)^2}$
- 7 What is $\int 4\sqrt{2x-3} \, dx$?
- A. $\frac{1}{3}(\sqrt{2x-3})^3 + c$
- B. $\frac{3}{4}(\sqrt{2x-3})^3 + c$
- C. $\frac{1}{2\sqrt{2x-3}} + c$
- D. $\frac{4}{3}(\sqrt{2x-3})^3 + c$
- 8 What transformation is required to produce the graph of $y = 3^{x+1}$ from the graph of $y = 3^x$?
- A. Shift left by 3 units
- B. Shift right by 1 unit
- C. Dilate vertically by a factor of 3
- D. Dilate horizontally by a factor of 3

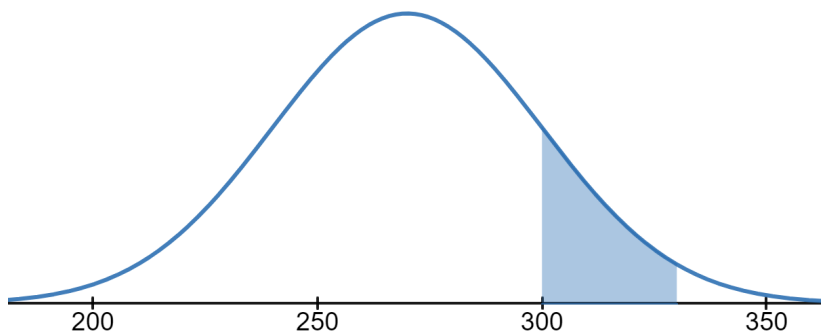
- 9 The heights of trees in a street are normally distributed with a mean of 270 cm and standard deviation of 30 cm. A particular tree is taller than 92% of the trees in the same street.

Which shaded region most accurately represents where the height of this tree lies?

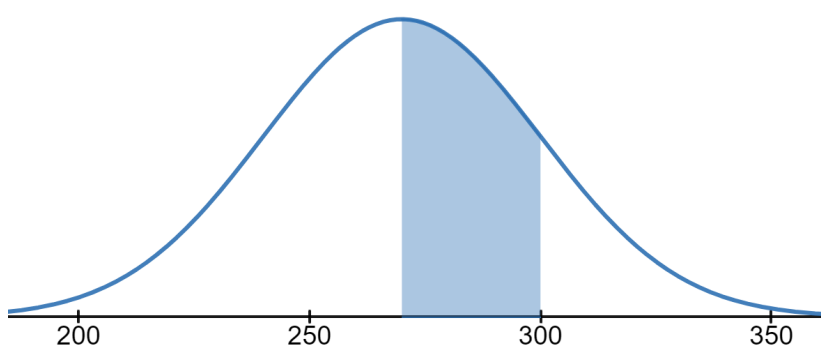
A.



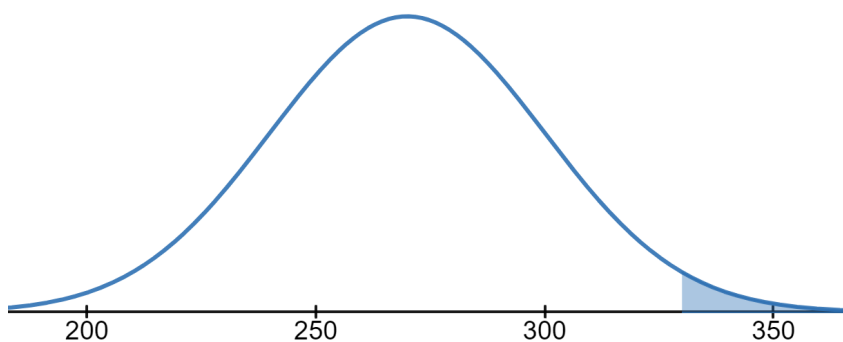
B.



C.



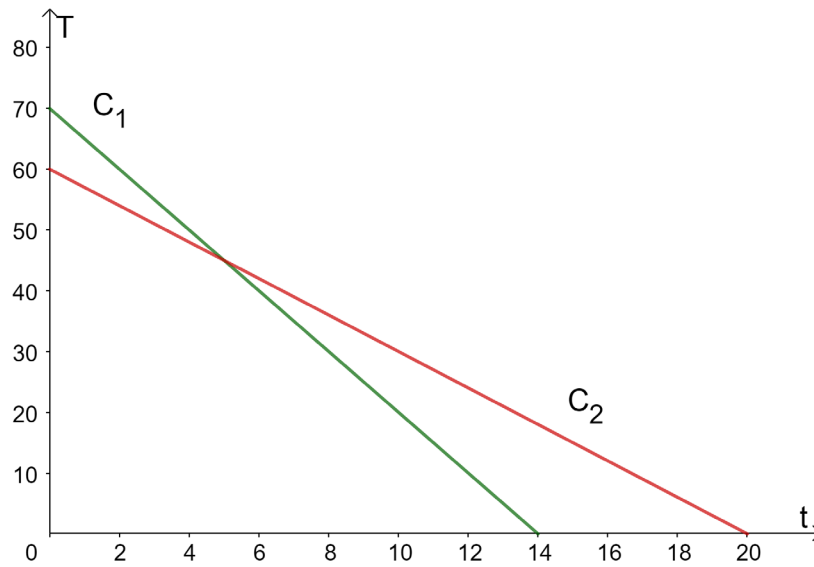
D.



- 10 Which of the following represents the domain of the function $f(x) = \frac{1}{\ln(x-2)}$?
- A. $[2, \infty)$
 - B. $(2, \infty)$
 - C. $(2, 3) \cup (3, \infty)$
 - D. $(3, \infty)$

Question 11 (4 marks)

The diagram shows the temperatures of two containers, C_1 and C_2 , after t hours of the initial measurement.



- (a) State the linear model for the temperature of each container.

2

- (b) For how many minutes will C_2 be cooler than C_1 ?

2

Question 12 (2 marks)

Solve $|3x - 5| = \frac{1}{2}$.

2

Question 13 (3 marks)

(a) Differentiate e^{x^2} .

1

(b) Hence, evaluate $\int_0^2 xe^{x^2} dx$.

2

Question 14 (4 marks)

For a sample of twelve students studying a certain course, the number of absent days and their final exam scores are recorded.

<i>Absent Days</i>	<i>Exam Score</i>
0	92
0	76
1	87
2	90
2	83
3	83
4	77
4	68
4	69
5	72
6	66
7	58

- (a) Determine the equation of the least-squares regression line, $y = mx + b$, for this data. State the coefficients correct to three decimal places.

1

- (b) State the gradient of the regression line and interpret the value of this gradient in the given context.

2

- (c) Another student studying the same course was absent for 10 days.

1

Calculate the predicted exam score for this student using your answer to part (a). Give your answer correct to the nearest whole number.

Question 15 (2 marks)

The curve $y = f(x)$ passes through the point $(0,1)$ such that $f'(x) = \frac{8x}{2x^2 + 1}$.

2

Find its equation.

Question 16 (3 marks)

Sketch the graph of $y = \frac{4}{x+2} - 1$, labelling the asymptotes and axes intercepts.

3

Question 17 (3 marks)

Let $0 < x < \frac{\pi}{2}$.

3

Explain why the limiting sum $1 + \sin^2 x + \sin^4 x + \sin^6 x + \dots$ exists.

Hence, find the limiting sum in terms of x .

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Question 18 (3 marks)

A table for the present value for an annuity of \$1 is shown.

The present value of an annuity of \$1

Period	1%	2%	4%	6%	8%	10%
1	0.9901	0.9804	0.9615	0.9434	0.9259	0.9091
2	1.9704	1.9416	1.8861	1.8334	1.7833	1.7355
3	2.941	2.8839	2.7751	2.673	2.5771	2.4869
4	3.902	3.8077	3.6299	3.4651	3.3121	3.1699
5	4.8534	4.7135	4.4518	4.2124	3.9927	3.7908
6	5.7955	5.6014	5.2421	4.9173	4.6229	4.3553

- (a) What would be the present value of a \$5000 per year annuity at 6% per annum for six years, with interest compounding annually? Give your answer correct to the nearest dollar.

1

- (b) An annuity of \$2000 per quarter is invested at 8% per annum, compounded quarterly for one year.

1

What is the present value, correct to the nearest dollar, of the annuity?

- (c) An annuity has a present value of \$46 000.

1

What is the amount of each yearly payment, compounded at 8% per annum, that is required to accumulate this annuity after five years? Give your answer correct to the nearest dollar.

Question 19 (3 marks)

Describe the three transformations which, when applied in succession, change the graph of $y = 2 \sin x$ to the graph with equation $y = \sin\left(\frac{x + \pi}{3}\right)$.

3

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Question 20 (2 marks)

If $\int_0^8 f(x) dx = -7$ and $\int_0^4 f(x) = 6$, find the value of $\int_8^4 f(x) dx$.

2

[illegible]

Question 21 (4 marks)

In a marathon, numbered flags are placed at regular intervals along the course. Flag number 1 is 800 metres from the start and all the rest are 500 metres apart.

- (a) Show that the formula for the distance, F_n , that a competitor must run to reach flag number n is $F_n = 500n + 300$. **1**

- (b) How many flags are needed if the length of the course is 10 000 m? **1**

- (c) A runner retires from the race after completing 6000 m. **2**

Find the nearest flag number and the runner's distance from that flag when he retired.

Question 22 (6 marks)

On any given day, the number X of times Alexa waits at a red light while driving to work is a random variable with probability distribution given by

x	0	1	2	3
$P(X = x)$	a^2	$\frac{3a^2}{2}$	$\frac{7-32a}{6}$	$4a$

- (a) Show that $a = \frac{1}{5}$.

3

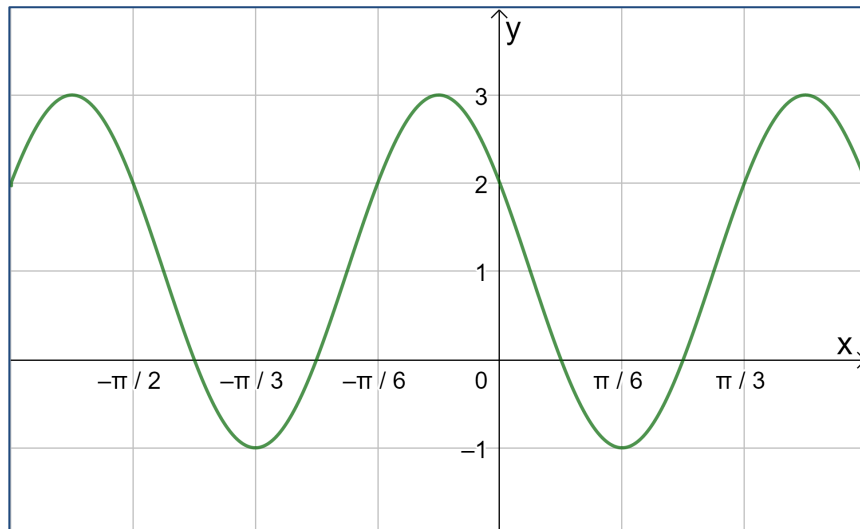
- (b) Let μ be the expected value of X .

3

Find $P(X < \mu)$.

Question 23 (5 marks)

The diagram shows the graph of $y = a \cos(bx + c) + d$.



(a) State the range.

1

(b) State the period.

1

(c) Find the values of a , b , c and d .

3

Question 24 (4 marks)

The lifespan in weeks, X , of a certain butterfly species is given by a probability density function.

$$f(x) = \begin{cases} \frac{\pi}{8} \sin\left(\frac{\pi x}{4}\right) & \text{for } 0 \leq x \leq 4 \\ 0 & \text{for all other values of } x \end{cases}$$

- (a) In a sample of 100 butterflies, how many are expected to live longer than three weeks? Give your answer correct to the nearest integer.

2

- (b) It is given that a butterfly has lived for more than two weeks.

2

What is the probability that the same butterfly lives for more than three weeks? Give your answer to two decimal places.

Question 25 (3 marks)

A random variable Z is normally distributed with mean 0 and standard deviation 1.

3

The table gives the probability of this random variable being greater than different values of z .

z	0	0.25	0.5	0.75	1	1.25	1.5	1.75	2
$P(Z > z)$	0.5	0.4013	0.3085	0.2266	0.1587	0.1057	0.668	0.0401	0.0228

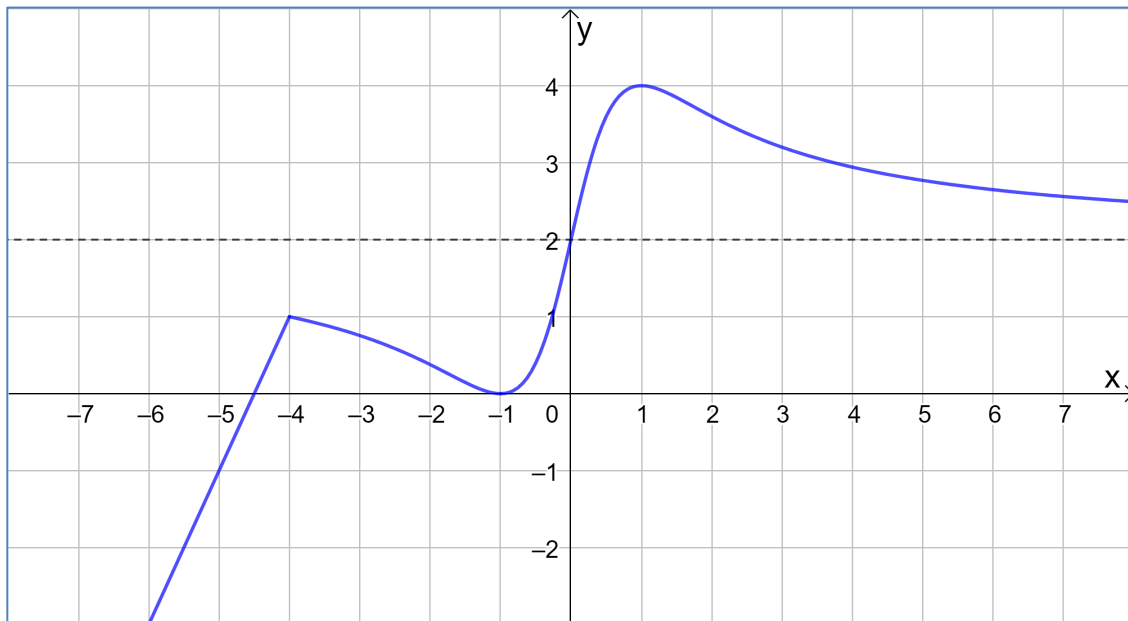
In a hospital of 3125 patients, it is found that 708 patients have a body temperature higher than $36.8^{\circ}C$. The body temperatures are distributed normally with a mean μ and standard deviation $0.3^{\circ}C$.

Using the given table, find the value of μ , correct to two decimal places.

[illegible]

Question 26 (6 marks)

The diagram shows the graph of $y = f(x)$ which is represented by a line for $x \leq -4$. The graph has stationary points at $x = -1$, $x = 1$ and approaches the horizontal asymptote $y = 2$ along the positive x -axis.



- (a) State the value of x for which $f'(x)$ does not exist.

1

- (b) Describe the behaviour of $f'(x)$ as x approaches infinity.

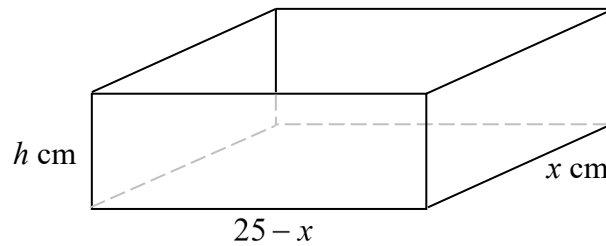
1

- (c) Sketch the graph of $y = f'(x)$ on the same diagram above.

4

Question 27 (6 marks)

The base of an open topped toy box has length $25 - x$ cm and width x cm. It has a total external surface area of 300 cm^2 .



- (a) Show that the height h of the toy box is given by $h = \frac{x^2 - 25x + 300}{50}$.

2

[illegible]

- (b) Find the dimensions of the base which give the maximum volume.

4

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Question 28 (8 marks)

Consider the function $f(x) = e^{2x} - 4e^x - 5$.

(a) Solve $f(x) = 0$.

2

(b) Find the coordinates of the stationary point.

3

(c) Find the range of the function.

1

Question 28 continues on page 23

Question 28 (continued)

(d) Find the interval for which the graph of the function is concave down.

2

End of Question 28

Question 29 (5 marks)

On his date of retirement Alex has \$650 000 in his superannuation fund.

5

His fund is earning interest at a rate of 0.5% per month. Each month, immediately after the interest has been paid, Alex withdraws \$5000.

How many years, correct to one decimal place, will the fund last?

[illegible]

Question 30 (6 marks)

The concentration of a chemical in a container changes according to the formula $C(t) = te^{-\frac{t}{10}}$, where t is time in minutes.

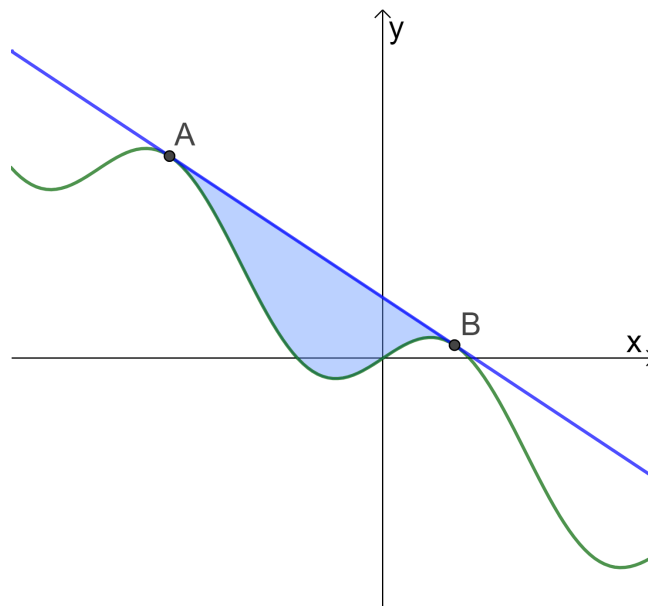
- (a) After how many minutes will the concentration reach its maximum? **2**

- (b) Find the time when the concentration of the chemical decreases at the fastest rate. **3**

- (c) Find the maximum rate of increase. **1**

Question 31 (5 marks)

In the diagram, the line passing through the points A and B is tangent to the graph of $y = \sin 2x - x$ at A and B . The x -coordinate of point B is $\frac{\pi}{4}$. 5



Find the exact area of the shaded region.

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Question 32 (3 marks)

Simplify $a^{\frac{\log_b c}{1+\log_b\left(\frac{a}{b}\right)}}$.

3

[illegible]

Section II extra writing space

If you use this space, clearly indicate which question you are answering.

[illegible]

Section II extra writing space

If you use this space, clearly indicate which question you are answering.

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

STUDENT ID _____

2022 Mathematics Advanced Yr 12
Section I Answer Sheet

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Select the alternative A, B, C or D that best answers the question. Fill in the response circle completely.

- | | | | | |
|-----------|-------------------------|-------------------------|-------------------------|-------------------------|
| 1 | A ☐ | B ☐ | C ☐ | D ☐ |
| 2 | A ☐ | B ☐ | C ☐ | D ☐ |
| 3 | A ☐ | B ☐ | C ☐ | D ☐ |
| 4 | A ☐ | B ☐ | C ☐ | D ☐ |
| 5 | A ☐ | B ☐ | C ☐ | D ☐ |
| 6 | A ☐ | B ☐ | C ☐ | D ☐ |
| 7 | A ☐ | B ☐ | C ☐ | D ☐ |
| 8 | A ☐ | B ☐ | C ☐ | D ☐ |
| 9 | A ☐ | B ☐ | C ☐ | D ☐ |
| 10 | A ☐ | B ☐ | C ☐ | D ☐ |

Mathematics Advanced
Mathematics Extension 1
Mathematics Extension 2

REFERENCE SHEET

Measurement

Length

$$l = \frac{\theta}{360} \times 2\pi r$$

Area

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a+b)$$

Surface area

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

Volume

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Functions

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For $ax^3 + bx^2 + cx + d = 0$:

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\text{and } \alpha\beta\gamma = -\frac{d}{a}$$

Relations

$$(x-h)^2 + (y-k)^2 = r^2$$

Financial Mathematics

$$A = P(1+r)^n$$

Sequences and series

$$T_n = a + (n-1)d$$

$$S_n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}(a+l)$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(r^n - 1)}{r - 1} = \frac{a(1 - r^n)}{1 - r}, r \neq 1$$

$$S = \frac{a}{1-r}, |r| < 1$$

Logarithmic and Exponential Functions

$$\log_a a^x = x = a^{\log_a x}$$

$$\log_a x = \frac{\log_b x}{\log_b a}$$

$$a^x = e^{x \ln a}$$

Trigonometric Functions

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2}ab \sin C$$

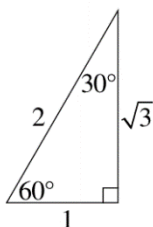
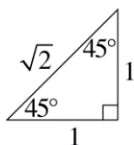
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$l = r\theta$$

$$A = \frac{1}{2}r^2\theta$$



Trigonometric identities

$$\sec A = \frac{1}{\cos A}, \quad \cos A \neq 0$$

$$\operatorname{cosec} A = \frac{1}{\sin A}, \quad \sin A \neq 0$$

$$\cot A = \frac{\cos A}{\sin A}, \quad \sin A \neq 0$$

$$\cos^2 x + \sin^2 x = 1$$

Compound angles

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\text{If } t = \tan \frac{A}{2} \text{ then } \sin A = \frac{2t}{1+t^2}$$

$$\cos A = \frac{1-t^2}{1+t^2}$$

$$\tan A = \frac{2t}{1-t^2}$$

$$\cos A \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)]$$

$$\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)]$$

$$\sin A \cos B = \frac{1}{2} [\sin(A + B) + \sin(A - B)]$$

$$\cos A \sin B = \frac{1}{2} [\sin(A + B) - \sin(A - B)]$$

$$\sin^2 nx = \frac{1}{2} (1 - \cos 2nx)$$

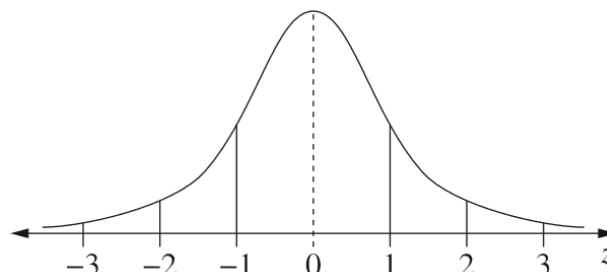
$$\cos^2 nx = \frac{1}{2} (1 + \cos 2nx)$$

Statistical Analysis

$$z = \frac{x - \mu}{\sigma}$$

An outlier is a score
less than $Q_1 - 1.5 \times IQR$
or
more than $Q_3 + 1.5 \times IQR$

Normal distribution



- approximately 68% of scores have z-scores between -1 and 1
- approximately 95% of scores have z-scores between -2 and 2
- approximately 99.7% of scores have z-scores between -3 and 3

$$E(X) = \mu$$

$$\operatorname{Var}(X) = E[(X - \mu)^2] = E(X^2) - \mu^2$$

Probability

$$P(A \cap B) = P(A)P(B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) \neq 0$$

Continuous random variables

$$P(X \leq x) = \int_a^x f(x) dx$$

$$P(a < X < b) = \int_a^b f(x) dx$$

Binomial distribution

$$P(X = r) = {}^nC_r p^r (1-p)^{n-r}$$

$$X \sim \operatorname{Bin}(n, p)$$

$$\Rightarrow P(X = x)$$

$$= \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, \dots, n$$

$$E(X) = np$$

$$\operatorname{Var}(X) = np(1-p)$$

Differential Calculus**Function****Derivative**

$$y = f(x)^n \quad \frac{dy}{dx} = nf'(x)[f(x)]^{n-1}$$

$$y = uv \quad \frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$y = g(u), u = f(x) \quad \frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$y = \frac{u}{v} \quad \frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$y = \sin f(x) \quad \frac{dy}{dx} = f'(x) \cos f(x)$$

$$y = \cos f(x) \quad \frac{dy}{dx} = -f'(x) \sin f(x)$$

$$y = \tan f(x) \quad \frac{dy}{dx} = f'(x) \sec^2 f(x)$$

$$y = e^{f(x)} \quad \frac{dy}{dx} = f'(x) e^{f(x)}$$

$$y = \ln f(x) \quad \frac{dy}{dx} = \frac{f'(x)}{f(x)}$$

$$y = a^{f(x)}$$

$$\frac{dy}{dx} = (\ln a) f'(x) a^{f(x)}$$

$$y = \log_a f(x) \quad \frac{dy}{dx} = \frac{f'(x)}{(\ln a) f(x)}$$

$$y = \sin^{-1} f(x) \quad \frac{dy}{dx} = \frac{f'(x)}{\sqrt{1-[f(x)]^2}}$$

$$y = \cos^{-1} f(x) \quad \frac{dy}{dx} = -\frac{f'(x)}{\sqrt{1-[f(x)]^2}}$$

$$y = \tan^{-1} f(x) \quad \frac{dy}{dx} = \frac{f'(x)}{1+[f(x)]^2}$$

Integral Calculus

$$\int f'(x)[f(x)]^n dx = \frac{1}{n+1}[f(x)]^{n+1} + c$$

where $n \neq -1$

$$\int f'(x) \sin f(x) dx = -\cos f(x) + c$$

$$\int f'(x) \cos f(x) dx = \sin f(x) + c$$

$$\int f'(x) \sec^2 f(x) dx = \tan f(x) + c$$

$$\int f'(x) e^{f(x)} dx = e^{f(x)} + c$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + c$$

$$\int f'(x) a^{f(x)} dx = \frac{a^{f(x)}}{\ln a} + c$$

$$\int \frac{f'(x)}{\sqrt{a^2 - [f(x)]^2}} dx = \sin^{-1} \frac{f(x)}{a} + c$$

$$\int \frac{f'(x)}{a^2 + [f(x)]^2} dx = \frac{1}{a} \tan^{-1} \frac{f(x)}{a} + c$$

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

$$\int_a^b f(x) dx \approx \frac{b-a}{2n} \{f(a) + f(b) + 2[f(x_1) + \dots + f(x_{n-1})]\}$$

where $a = x_0$ and $b = x_n$

Combinatorics

$${}^nP_r = \frac{n!}{(n-r)!}$$

$$\binom{n}{r} = {}^nC_r = \frac{n!}{r!(n-r)!}$$

$$(x+a)^n = x^n + \binom{n}{1}x^{n-1}a + \dots + \binom{n}{r}x^{n-r}a^r + \dots + a^n$$

Vectors

$$|\underline{u}| = |x\underline{i} + y\underline{j}| = \sqrt{x^2 + y^2}$$

$$\underline{u} \cdot \underline{v} = |\underline{u}||\underline{v}|\cos\theta = x_1x_2 + y_1y_2,$$

$$\text{where } \underline{u} = x_1\underline{i} + y_1\underline{j}$$

$$\text{and } \underline{v} = x_2\underline{i} + y_2\underline{j}$$

$$\underline{r} = \underline{a} + \lambda\underline{b}$$

Complex Numbers

$$z = a + ib = r(\cos\theta + i\sin\theta)$$

$$= re^{i\theta}$$

$$[r(\cos\theta + i\sin\theta)]^n = r^n(\cos n\theta + i\sin n\theta)$$

$$= r^n e^{in\theta}$$

Mechanics

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

$$x = a \cos(nt + \alpha) + c$$

$$x = a \sin(nt + \alpha) + c$$

$$\ddot{x} = -n^2(x - c)$$

2022 Mathematics Advanced Year 12 Marking Guidelines

Section 1

Multiple-choice Answer Key

Question	Answer
1	B
2	C
3	B
4	D
5	A
6	A
7	D
8	C
9	B
10	C

Please see multiple-choice explanations over.

Question	Solution
1	$a_1 = 12$ $a_{15} = 67$ $a_{15} = a_1 + 14d \Rightarrow d = \frac{55}{14}$
2	centre $\Rightarrow (x-1)^2 + (y+2)^2 = r^2$ $r = \sqrt{(-2-0)^2 + (1-0)^2} = \sqrt{5}$
3	$f'(x) = 12x^2 - 12x + 3$ $f'(x) = 0 \Rightarrow 3(4x^2 - 4x + 1) = 0 \Rightarrow 3(2x-1)^2 = 0$ $x = \frac{1}{2}$
4	$(1 + \tan^2 x) - (\sin^2 x + \cos^2 x) = 1 + \tan^2 x - 1 = \tan^2 x$
5	no x-intercept $\Rightarrow \Delta < 0 \Rightarrow b^2 - 8 < 0$ x coordinate of turning point $< 0 \Rightarrow -\frac{b}{2} < 0$ among the choices only $b = 2$ satisfy these conditions
6	$\frac{d}{dx} 2(1-e^x)^{-1} = -2(1-e^x)^{-2} \times -e^x = \frac{2e^x}{(1-e^x)^2}$
7	$\int 4(2x-3)^{\frac{1}{2}} dx = \frac{4(2x-3)^{\frac{3}{2}}}{\frac{3}{2} \times 2} + c = \frac{4}{3}(\sqrt{2x-3})^3 + c$
8	$3^{x+1} = 3 \times 3^x$
9	Height is between 1 to 2 standard deviations above the mean $270 + 30 < h < 270 + 60 \Rightarrow 300 < h < 330$
10	$\begin{cases} x-2 > 0 \\ \ln(x-2) \neq 0 \end{cases} \Rightarrow \begin{cases} x > 2 \\ x \neq 3 \end{cases}$

Section II

Note: An incorrect answer in a previous part will not necessarily preclude students from achieving full marks in a later part. Answers here are based on correct prior part answers. Marking will need to adapt to pursue correct method with the use of incorrect prior parts.

Question 11 (a)

Criteria	Marks
• Provides two correct equations	2
• Provides one correct equation or two correct gradients	1

Sample answer:

$$C_1 = -5t + 70$$

$$C_2 = -3t + 60$$

Question 11 (b)

Criteria	Marks
• Provides the correct answer	2
• Provides $t < 5$ or $t = 5$	1

Sample answer:

$$C_2 < C_1 \Rightarrow -3t + 60 < -5t + 70 \Rightarrow 2t < 10 \Rightarrow t < 5$$

$$5 \times 60 = 300 \text{ minutes}$$

Question 12

Criteria	Marks
• Provides the correct answer	2
• Provides one correct solution	1

Sample answer:

$$\begin{cases} 3x - 5 = \frac{1}{2} \\ 3x - 5 = -\frac{1}{2} \end{cases} \Rightarrow \begin{cases} x = \frac{11}{6} \\ x = \frac{3}{2} \end{cases}$$

Question 13 (a)

Criteria	Marks
Provides the correct answer	1

Sample answer:

$$2xe^{x^2}$$

Question 13 (b)

Criteria	Marks
Provides the correct answer	2
Provides a correct antiderivative	1

Sample answer:

$$\int_0^2 xe^{x^2} dx = \left[\frac{1}{2} e^{x^2} \right]_0^2 = \frac{e^4 - 1}{2}$$

Question 14 (a)

Criteria	Marks
Provides the correct answer	1

Sample answer:

$$y = -3.889x + 89.066$$

Question 14 (b)

Criteria	Marks
Provides the correct answer	2
Provides either the correct gradient or the interpretation	1

Sample answer:

$$\text{gradient} = -3.889$$

more absenteeism will result in less exam score

Question 14 (c)

Criteria	Marks
Provides the correct answer	1

Sample answer:

$$-3.889 \times 10 + 89.066 = 50 \text{ (nearest integer)}$$

Question 15

Criteria	Marks
• Provides the correct answer	2
• Provides one correct solution	1

Sample answer:

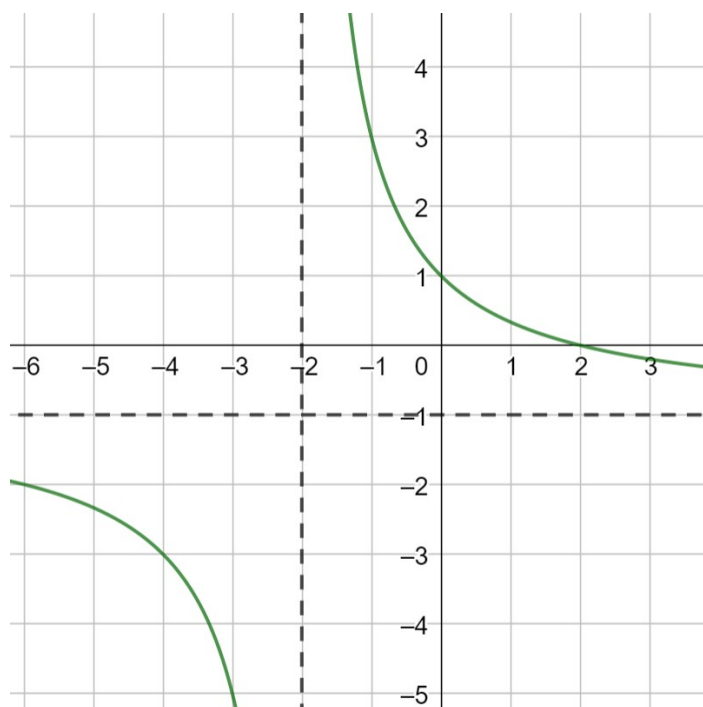
$$\int \frac{8x}{2x^2 + 1} dx = 2 \ln(2x^2 + 1) + c$$

$$2 \ln 1 + c = 1 \Rightarrow c = 1$$

$$f(x) = 2 \ln(2x^2 + 1) + 1$$

Question 16

Criteria	Marks
• Provides the correct sketch with the asymptotes and axis intercepts labelled	3
• Provides the correct sketch	2
• Provides the correct asymptotes or axis intercepts	1

Sample answer:Asymptotes: $x = -2$, $y = -1$ Intercepts: $(2, 0)$, $(0, 1)$ 

Question 17

Criteria	Marks
• Provides the correct answer including justification why the sum exists	3
• Provides the correct common ratio and justifies why the sum exists	2
• Provides the correct common ratio	1

Sample answer:

$$r = \sin^2 x$$

$$0 < r < 1 \text{ since } 0 < x < \frac{\pi}{2}$$

$\therefore$ sum exists

$$\text{Sum} = \frac{1}{1 - \sin^2 x} = \frac{1}{\cos^2 x} \text{ or } \sec^2 x$$

Question 18 (a)

Criteria	Marks
• Provides the correct answer	1

Sample answer:

$$\$5000 \times 4.9173 = \$24587 \text{ (nearest \$)}$$

Question 18 (b)

Criteria	Marks
• Provides the correct answer	1

Sample answer:

$$\$2000 \times 3.8077 = \$7615 \text{ (nearest \$)}$$

Question 18 (c)

Criteria	Marks
• Provides the correct answer	1

Sample answer:

$$x \times 3.9927 = \$46000$$

$$x = \$11521 \text{ (nearest \$)}$$

Question 19

Criteria	Marks
• Provides the correct answer	3
• Provides two correct transformations with correct order	2
• Provides one correct transformation	1

Sample answer:

$$2 \sin x \rightarrow \sin x \rightarrow \sin\left(\frac{1}{3}x\right) \rightarrow \sin\left(\frac{1}{3}(x + \pi)\right)$$

vertical dilation by factor $\frac{1}{2}$

horizontal dilation by factor 3

translation π units left**Question 20**

Criteria	Marks
• Provides the correct answer	2
• Provides $\int_0^8 f(x) dx$ as sum of two definite integrals including 4 as one boundary	1

Sample answer:

$$\int_0^8 f(x) dx = \int_0^4 f(x) dx + \int_4^8 f(x) dx = \int_0^4 f(x) dx - \int_8^4 f(x) dx$$

$$\int_8^4 f(x) dx = \int_0^4 f(x) dx - \int_0^8 f(x) dx = 6 - (-7) = 13$$

Question 21 (a)

Criteria	Marks
Provides the correct solution	1

Sample answer:

$$F_n = F_1 + (n-1)d = 800 + (n-1)500 = 500n + 300$$

Question 21 (b)

Criteria	Marks
Provides the correct answer	1

Sample answer:

$$500n + 300 = 10000$$

$$n = 19.4$$

19 flags

Question 21 (c)

Criteria	Marks
Provides the correct answer	2
Provides the correct flag number	1

Sample answer:

$$500n + 300 = 6000$$

$$n = 11.4$$

$$0.4 \times 500 = 200$$

200 metres from the 11th flag

Question 22 (a)

Criteria	Marks
• Provides the correct answer with justification	3
• Provides the correct solutions to the quadratic equation	2
• Provides an equivalent of the correct quadratic equation	1

Sample answer:

$$a^2 + \frac{3a^2}{2} + \frac{7-32a}{6} + 4a = 1$$

$$\frac{15a^2 - 8a + 7}{6} = 0$$

$$a = \frac{1}{5} \text{ or } a = \frac{1}{3}$$

$$a \neq \frac{1}{3} \text{ since } P(X=3) = \frac{4}{3} > 1$$

$$\therefore a = \frac{1}{5} \text{ for which } 0 \leq P(X=x) \leq 1 \text{ for all } x$$

Question 22 (b)

Criteria	Marks
• Provides the correct answer	3
• Provides the correct expected value	2
• Provides the correct expression leading to the expected value	1

Sample answer:

$$\mu = 0 \times \left(\frac{1}{5}\right)^2 + 1 \times \frac{3\left(\frac{1}{5}\right)^2}{2} + 2 \times \frac{7-32\left(\frac{1}{5}\right)}{6} + 3 \times 4\left(\frac{1}{5}\right) = \frac{131}{50} \approx 2.62$$

$$P\left(X < \frac{131}{50}\right) = 1 - P(X=3) = 1 - \frac{4}{5} = \frac{1}{5}$$

Question 23 (a)

Criteria	Marks
Provides the correct answer	1

Sample answer: $[-1, 3]$ **Question 23 (b)**

Criteria	Marks
Provides the correct answer	1

Sample answer:

$$\frac{\pi}{6} - \left(-\frac{\pi}{3}\right) = \frac{\pi}{2}$$

Question 23 (c)

Criteria	Marks
Provides the correct answer	3
Provides three correct values	2
Provides two correct values	1

Sample answer: $d = 1$ (axis of oscillation) $a = 2$ (amplitude)

$$\frac{2\pi}{b} = \frac{\pi}{2} \Rightarrow b = 4 \text{ (period)}$$

 $(0, 2)$ is on the graph of $y = 2 \cos(4x + c) + 1$

$$2 \cos(c) + 1 = 2 \Rightarrow \cos(c) = \frac{1}{2} \Rightarrow c = \frac{\pi}{3}$$

Question 24 (a)

Criteria	Marks
• Provides the correct answer	2
• Provides correct antiderivative with endpoints	1

Sample answer:

$$P(X > 3) = \int_3^4 \frac{\pi}{8} \sin\left(\frac{\pi x}{4}\right) dx = \left[-\frac{1}{2} \cos\left(\frac{\pi x}{4}\right) \right]_3^4 = \frac{1}{2} - \frac{\sqrt{2}}{4} \approx 0.146$$

 $100 \times 0.146 \approx 15$ butterflies
Question 24 (b)

Criteria	Marks
• Provides the correct answer	2
• Provides correct value of the numerator or denominator for the conditional probability	1

Sample answer:

$$P(X > 3 | X > 2) = \frac{P(X > 3)}{P(X > 2)} = \frac{\frac{1}{2} - \frac{\sqrt{2}}{4}}{\int_2^4 \frac{\pi}{8} \sin\left(\frac{\pi x}{4}\right) dx} = \frac{\frac{2 - \sqrt{2}}{4}}{\frac{1}{2}} = \frac{2 - \sqrt{2}}{2} = 0.29 \text{ (2 dp)}$$

Question 25

Criteria	Marks
• Provides the correct answer	3
• Relates 36.8 with z-score	2
• Relates the working out with 0.2266	1

Sample answer:Let T be body temperature

$$P(T > 36.8) = \frac{708}{3125} \approx 0.2266$$

From the table, $P(Z > 0.75) = 0.2266$

$$\frac{36.8 - \mu}{0.3} = 0.75 \Rightarrow \mu = 36.58 \text{ (2 dp)}$$

Question 26 (a)

Criteria	Marks
Provides the correct answer	1

Sample answer:

$$x = -4$$

Question 26 (b)

Criteria	Marks
Provides the correct answer	1

Sample answer:

$$f'(x) \rightarrow 0$$

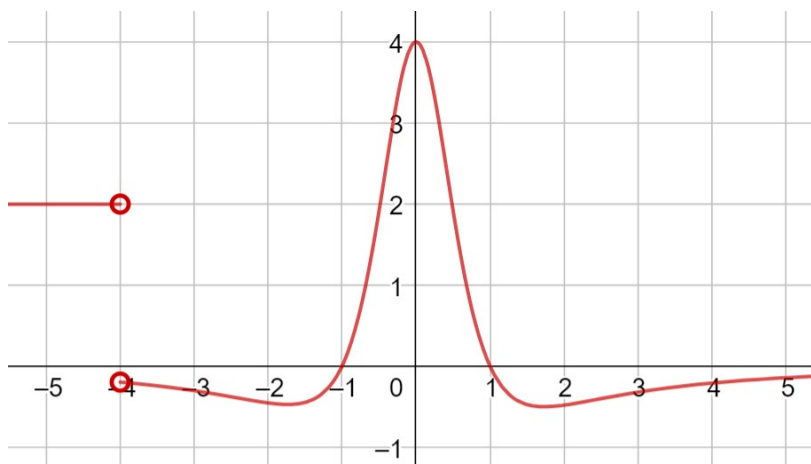
Question 26 (c)

Criteria	Marks
Provides the correct answer with all required features	4
Provides 5-6 of the required features	3
Provides 3-4 of the required features	2
Provides 1-2 of the required features	1

Sample answer:

Required features

- x -intercepts at -1 and 1
- $f'(x) = 2$ for $x < -4$
- Discontinuity at $x = -4$ and $f'(-4)$ is undefined
- $y = 0$ is an asymptote for $x \rightarrow \infty$
- $f'(x) > 0$ for $-1 < x < 1$ with a maximum point in the graph
- $f'(x) < 0$ for $-4 < x < -1$ with a minimum point in the graph
- $f'(x) < 0$ for $x > 1$ with a minimum point in the graph



Question 27 (a)

Criteria	Marks
• Provides the correct solution	2
• Provides the correct expression for the surface area in terms of x and h	1

Sample answer:

$$(25 - x)x + 2hx + 2h(25 - x) = 300$$

$$h(2x + 2(25 - x)) = 300 - (25 - x)x$$

$$h = \frac{300 - 25x + x^2}{2x + 50 - 2x} = \frac{x^2 - 25x + 300}{50}$$

Question 27 (b)

Criteria	Marks
• Provides the correct answer	4
• Provides the correct solutions for $V'(x) = 0$	3
• Provides an expression for $V'(x)$	2
• Provides the correct expression for the volume in terms of x (or x and h)	1

Sample answer:

$$\text{Volume} = (25 - x)xh = \frac{(25x - x^2)(x^2 - 25x + 300)}{50} = V(x)$$

$$V'(x) = \frac{1}{50}((25 - 2x)(x^2 - 25x + 300) + (25x - x^2)(2x - 25)) = \frac{1}{50}(25 - 2x)(x^2 - 25x + 300 + x^2 - 25x)$$

$$V'(x) = 0 \Rightarrow (25 - 2x)(2x^2 - 50x + 300) = 0$$

$$x = 10, x = 15, x = \frac{25}{2}$$

x	10	$\frac{25}{2}$	15
V(x)	450	449.219	450

Maximum volume will occur when base is 15 cm × 10 cm

Question 28 (a)

Criteria	Marks
• Provides the correct answer	2
• Attempts to solve the equation quadratic in form	1

Sample answer:

$$u = e^x \Rightarrow u^2 - 4u - 5 = 0 \Rightarrow \begin{cases} u = -1 \\ u = 5 \end{cases} \Rightarrow \begin{cases} e^x = -1 \\ e^x = 5 \end{cases} \Rightarrow x = \ln 5$$

Question 28 (b)

Criteria	Marks
• Provides the correct answer	3
• Provides the correct solution to $f'(x) = 0$	2
• Provides the correct derivative	1

Sample answer:

$$f'(x) = 0 \Rightarrow 2e^{2x} - 4e^x = 0 \Rightarrow 2e^x(e^x - 2) = 0 \Rightarrow x = \ln 2$$

$$f(\ln 2) = e^{2\ln 2} - 4e^{\ln 2} - 5 = 4 - 8 - 5 = -9 \Rightarrow (\ln 2, -9)$$

Question 28 (c)

Criteria	Marks
• Provides the correct answer	1

Sample answer:

$$\begin{cases} f'(\ln 4) > 0 \\ f'(\ln 1) < 0 \end{cases} \Rightarrow x = \ln 2 \text{ is a minimum}$$

$$\text{Range} = [-9, \infty)$$

Question 28 (d)

Criteria	Marks
• Provides the correct answer	2
• Provides the correct second derivative	1

Sample answer:

$$f''(x) < 0 \Rightarrow 4e^{2x} - 4e^x < 0 \Rightarrow 4e^x(e^x - 1) < 0 \Rightarrow e^x < 1 \Rightarrow x < 0$$

Question 29

Criteria	Marks
• Provides the correct answer	5
• Provides the value for r^n	4
• Provides A_n using sum of geometric series	3
• Provides a generalisation for A_n	2
• Provides the correct expression for A_1	1

Sample answer:

Let A_n be the value of the fund after the n -th withdrawal.

$$M = 650000, r = 1 + 0.005, W = 5000$$

$$A_1 = Mr - W$$

$$A_2 = A_1r - W = Mr^2 - Wr - W$$

$$A_3 = A_2r - W = Mr^3 - Wr^2 - Wr - W$$

$$A_n = Mr^n - W(r^{n-1} + r^{n-2} + \dots + r + 1) = Mr^n - W \frac{r^n - 1}{r - 1}$$

When the fund expires, $A_n = 0$

$$Mr^n - W \frac{r^n - 1}{r - 1} = 0 \Rightarrow 650000r^n - \frac{5000}{0.005}(r^n - 1) = 0$$

$$650000r^n - 1000000r^n + 1000000 = 0$$

$$r^n = \frac{1000000}{350000} \Rightarrow n = \frac{\ln\left(\frac{20}{7}\right)}{\ln 1.005} = 210.489 \dots \text{ months} = 17.5 \text{ years (1 d.p.)}$$

Question 30 (a)

Criteria	Marks
• Provides the correct answer	2
• Provides the correct derivative	1

Sample answer:

$$C'(t) = e^{-\frac{t}{10}} - \frac{t}{10} e^{-\frac{t}{10}}$$

$$C'(t) = 0 \Rightarrow e^{-\frac{t}{10}} \left(1 - \frac{t}{10}\right) = 0 \Rightarrow t = 10$$

$$\begin{cases} C'(9) > 0 \\ C'(11) < 0 \end{cases} \Rightarrow \text{max at } t = 10$$

Question 30 (b)

Criteria	Marks
• Provides the correct answer with justification	3
• Provides the correct value of t	2
• Provides the correct second derivative	1

Sample answer:

$$C''(t) = -\frac{1}{10}e^{-\frac{t}{10}} - \left(\frac{1}{10}e^{-\frac{t}{10}} - \frac{t}{100}e^{-\frac{t}{10}} \right) = e^{-\frac{t}{10}} \left(-\frac{2}{10} + \frac{t}{100} \right)$$

$$C''(t) = 0 \Rightarrow t = 20$$

$$\begin{cases} C''(19) < 0 \\ C''(21) > 0 \end{cases} \Rightarrow \text{min (decrease at fastest rate) at } t = 20$$

Question 30 (c)

Criteria	Marks
• Provides the correct answer	1

Sample answer:

$$t = 0$$

$$C''(t) = 0 \text{ only at } t = 20.$$

However, at $t = 20$, rate of decrease is maximum.

Therefore maximum rate of increase can only happen initially.

Question 31

Criteria	Marks
• Provides the correct answer	5
• Provides the correct antiderivative for the area with endpoints	4
• Provides the x-coordinate of point A	3
• Provides the equation of the tangent line	2
• Provides the gradient of the tangent line	1

Sample answer:

Equation of tangent

$$f'(x) = 2 \cos 2x - 1$$

$$m_{\text{tangent}} = f'\left(\frac{\pi}{4}\right) = -1$$

$$f\left(\frac{\pi}{4}\right) = 1 - \frac{\pi}{4}$$

$$\text{tangent line: } y - 1 + \frac{\pi}{4} = -1\left(x - \frac{\pi}{4}\right) \Rightarrow y = -x + 1$$

Point A

$$\sin 2x - x = -x + 1 \Rightarrow \sin 2x = 1$$

$$2x = \frac{\pi}{2} + 2\pi k \Rightarrow x = \frac{\pi}{4} + \pi k \Rightarrow x = -\frac{3\pi}{4}$$

$$\text{Area} = \int_{-\frac{3\pi}{4}}^{\frac{\pi}{4}} (-x + 1 - \sin 2x + x) dx = \left[x + \frac{\cos 2x}{2} \right]_{-\frac{3\pi}{4}}^{\frac{\pi}{4}} = \left(\frac{\pi}{4} + 0 \right) - \left(-\frac{3\pi}{4} + 0 \right) = \pi$$

Question 32

Criteria	Marks
• Provides the correct answer	3
• Applies at least two logarithm laws leading towards the solution	2
• Applies one logarithm law leading towards the solution	1

Sample answer:

$$\frac{\log_b c}{1 + \log_b \left(\frac{a}{b}\right)}$$

a

$$= a^{\frac{\log_b c}{\log_b b + \log_b \left(\frac{a}{b}\right)}}$$

$$= a^{\frac{\log_b c}{\log_b a}}$$

$$= a^{\log_a c}$$

$$= c$$

2022 Mathematics Advanced Year 12 Mapping Grid

Section I

Question	Marks	Content	Syllabus outcomes
1	1	MA-M1 Modelling Financial Situations	MA12-4
2	1	MA-F1 Working with Functions	MA11-8
3	1	MA-C3 Applications of Differentiation	MA12-6
4	1	MA-T2 Trigonometric Functions and Identities	MA11-4
5	1	MA-F1 Working with Functions	MA11-8
6	1	MA-C2 Differential Calculus	MA12-6
7	1	MA-C4 Integral Calculus	MA12-7
8	1	MA-F2: Graphing Techniques	MA12-1
9	1	MA-E1 Logarithms and Exponentials	MA11-8
10	1	MA-S3 Random Variables	MA12-8

Section II

Question	Marks	Content	Syllabus outcomes
11(a)	2	MA-F1 Working with Functions	MA11-2
11(b)	2	MA-F1 Working with Functions	MA11-2
12	2	MA-F1 Working with Functions	MA11-1
13(a)	1	MA-C2 Differential Calculus	MA12-6
13(b)	2	MA-C4 Integral Calculus	MA12-7
14(a)	1	MA-S2 Descriptive Statistics and Bivariate Data Analysis	MA12-9
14(b)	2	MA-S2 Descriptive Statistics and Bivariate Data Analysis	MA12-9
14(c)	1	MA-S2 Descriptive Statistics and Bivariate Data Analysis	MA12-9
15	2	MA-C4 Integral Calculus	MA12-7
16	3	MA-F1 Working with Functions	MA11-1
17	3	MA-M1 Modelling Financial Situations	MA12-4
18(a)	1	MA-M1 Modelling Financial Situations	MA12-9
18(b)	1	MA-M1 Modelling Financial Situations	MA12-9
18(c)	1	MA-M1 Modelling Financial Situations	MA12-9
19	3	MA-F2 Graphing Techniques	MA12-1
20	2	MA-C4 Integral Calculus	MA12-7
21(a)	1	MA-M1 Modelling Financial Situations	MA12-4
21(b)	1	MA-M1 Modelling Financial Situations	MA12-4
21(c)	2	MA-M1 Modelling Financial Situations	MA12-2
22(a)	3	MA-S1 Probability and Discrete Probability Distributions	MA11-7
22(b)	3	MA-S1 Probability and Discrete Probability Distributions	MA11-7
23(a)	1	MA-T3 Trigonometric Functions and Graphs	MA12-5

23(b)	1	MA-T3 Trigonometric Functions and Graphs	MA12-5
23(c)	3	MA-T3 Trigonometric Functions and Graphs	MA12-5
24(a)	2	MA-S3 Random Variables	MA12-8
24(b)	2	MA-S3 Random Variables	MA12-8
25	3	MA-S3 Random Variables	MA12-8
26(a)	1	MA-C1 Introduction to Differentiation	MA11-5
26(b)	1	MA-C1 Introduction to Differentiation	MA11-5
26(c)	4	MA-C1 Introduction to Differentiation	MA11-1
27(a)	2	MA-F1 Working with Functions	MA11-2
27(b)	4	MA-C3 Applications of Differentiation	MA12-2
28(a)	2	MA-E1 Logarithms and Exponentials	MA11-6
28(b)	3	MA-C3 Applications of Differentiation	MA12-6
28(c)	1	MA-F2 Graphing Techniques	MA12-2
28(d)	2	MA-C3 Applications of Differentiation	MA12-6
29	5	MA-M1 Modelling Financial Situations	MA12-2
30(a)	2	MA-C3 Applications of Differentiation	MA12-3
30(b)	3	MA-C3 Applications of Differentiation	MA12-3
30(c)	1	MA-C3 Applications of Differentiation	MA12-3
31	5	MA-C4 Integral Calculus	MA12-3
32	3	MA-E1 Logarithms and Exponentials	MA11-6